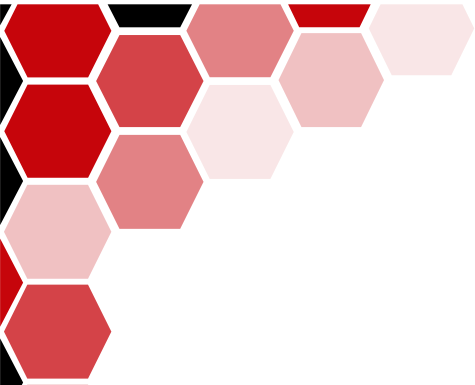




Lecture 1: **What, When, Where, and Why?**

ECE 340 – Electronic Circuits I Sec. 1

Shuva Mitra

09/03/2026

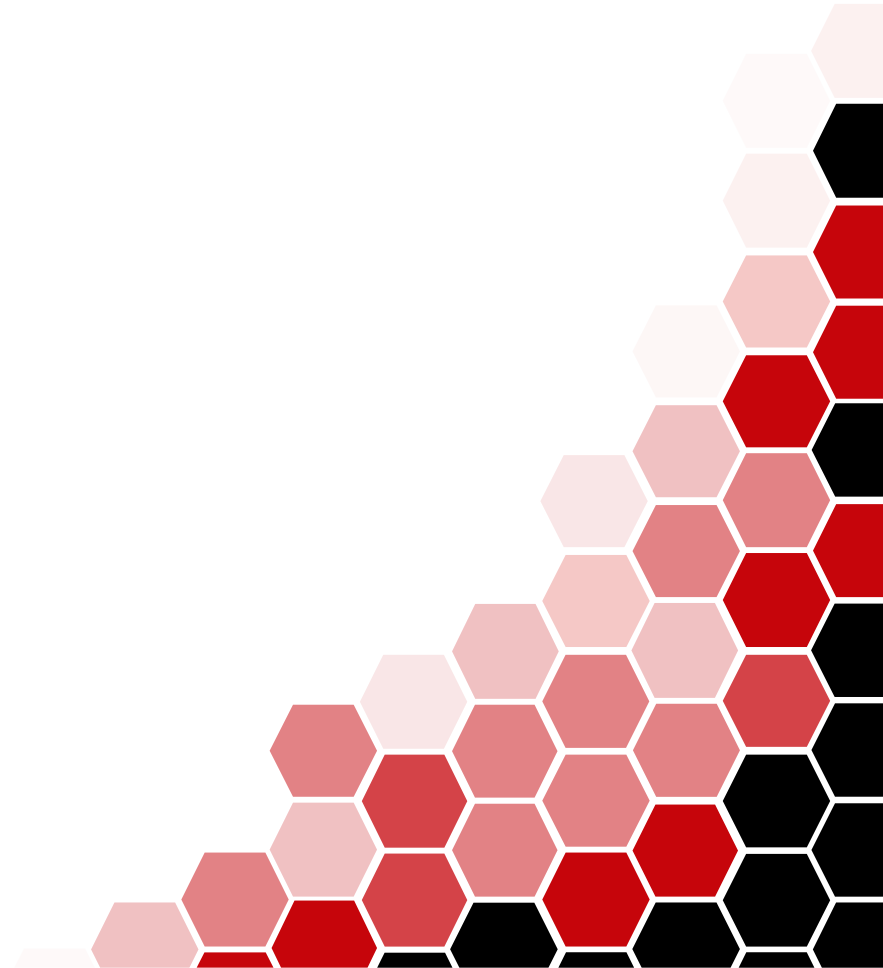


Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN-MADISON

Material credit: Prof. Jennifer Volk

Today's Outline

1. Course logistics
2. Signals intro
3. Recap – KCL/KVL & superposition
4. Thevenin & Norton theorems



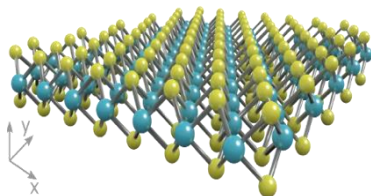
About me

Graduate student at *Quantum Transport Theory and Simulation Lab*

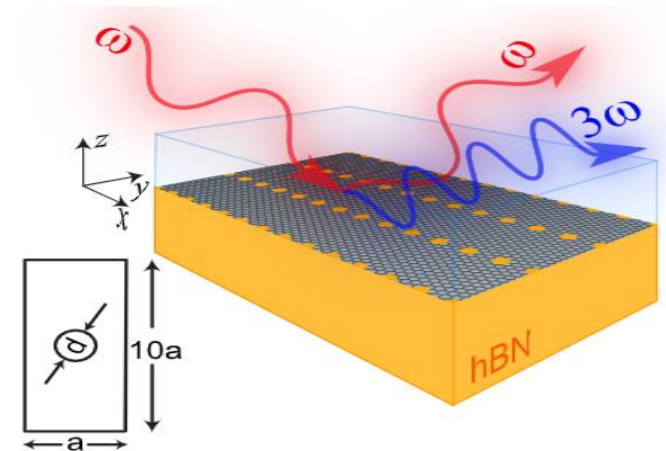
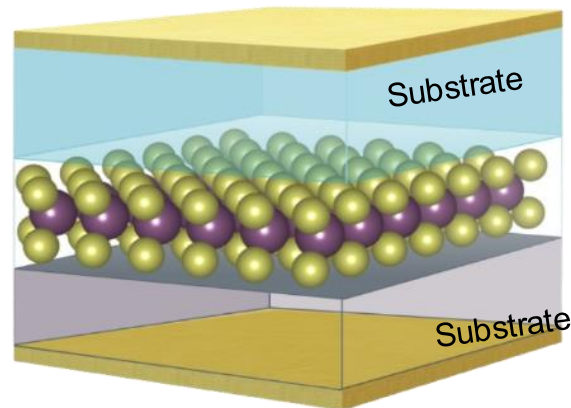
Advisor: Irena Knezevic

Research focus:

- Computational modeling of novel physics in semiconductors
- THz electron transport in nanoscale systems
- 2D Materials (Graphene, Transition-metal Dichalcogenides, etc.)



2D materials



Course logistics

Instructors and teaching assistants

Instructor title, name, e-mail

Lecturer (Student Assistant), Shuva Mitra, smitra8@wisc.edu

Instructor availability

Office hours:

Tuesday 5:20pm-6:20pm and Thursday 5:20pm-6:20pm in EH 3345 unless otherwise announced; otherwise by appointment.

Teaching assistant, e-mail, office hours

Daksh Sharma, dsharma45@wisc.edu

TA Office hours: Monday: 5:45 PM – 6:45 PM @ EH1422

Wednesday: 3:00 PM – 4:00 PM @ EH1422

Office hours are for homework help! Use them wisely.



Course logistics

Required textbook, software, & other course materials

- **Textbook: Adel S. Sedra, et al., “Microelectronic Circuits”, 6th edition, Oxford University Press, 2010.**
- Non-graphing scientific calculator that cannot store data. Used for homework, quizzes, and exams.

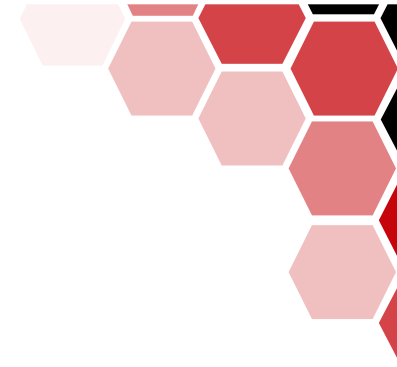
What will we cover?

Official course description: A first course in modeling, characterization, and application of semiconductor devices and integrated circuits. Development of appropriate models for circuit-level behavior of diodes, bi-polar and field effect transistors, and non-ideal op-amps. Application in analysis and design of linear amplifiers. Frequency domain characterization of transistor circuits.

What will we cover?

List of topics to be covered/textbook chapters

- **Chapter 1:** Signals and amplifiers
- **Chapter 3:** Semiconductors
- **Chapter 4:** Diodes
- **Chapter 5:** MOS field effect transistors (MOSFETs)
- **Chapter 6:** Bipolar junction transistors (BJTs)
- **Chapter 7:** Transistor amplifiers
- **Chapter 9:** Frequency response



What will we cover?

Course Learning Outcomes:

1. Analyze functions and DC and AC operations of diodes, MOSFETs, and BJTs
2. Utilize large and small signal models of diodes, MOSFETs, and BJTs in analysis of analog circuits
3. Analyze and design single stage analog amplifiers
4. Analyze and design basic differential amplifiers and multistage amplifiers
5. Analyze basic operational amplifier circuits

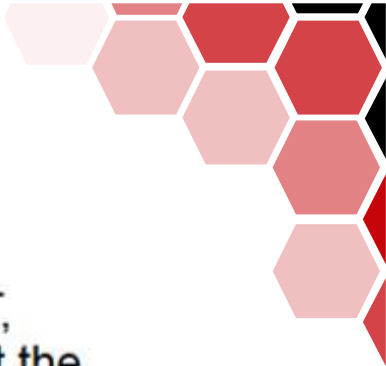
+ Suggested objective: explore this area beyond what the course offers
(read more literature, talk to me about research, ask questions)

Grading breakdown

- **Grading, exams, in-class assessments, homework, & other assignments**

90 and above	A
80-90	AB
70-80	B
60-70	BC
50-60	C
40-50	D

Grade assignment can change based on class performance. When the total score is between two grades, the higher grade will be assigned.



Use of Artificial Intelligence (AI) Tools:

We may use artificial intelligence (AI) tools and applications (such as ChatGPT, Claude, DALL-E, etc.) in some circumstances in this course when they support the course learning objectives. The specifics of when, where and how these tools are permitted will be included with each assignment, along with guidance for attribution. Any use of these tools other than where indicated is a violation of this course's expectations and will be addressed through UW–Madison's [academic misconduct policy](#), specifically UWS 14.03(1)b (b) Uses unauthorized materials or fabricated data in any academic exercise.

Considerations for Citing Generative AI can be found here:

<https://researchguides.library.wisc.edu/c.php?g=1334141&p=9825297>



Why is this useful?

Why is this useful?

Communication

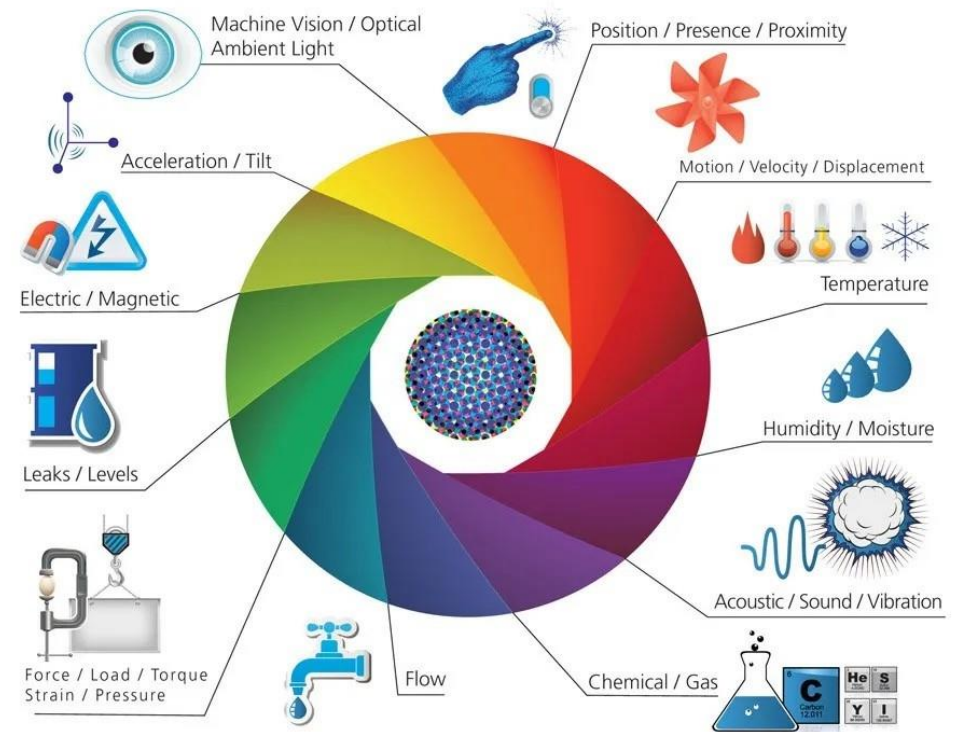


Power source regulation:

- Any electronics plugging into a wall
- Any electronics with a battery
- Any electronics with an engine



Sensing



It's an analog world and we're just discretizing in it!

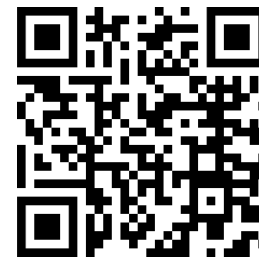
Why should I care?

Jack Kilby - Inventor of first IC



<https://www.youtube.com/watch?v=rF3FWHQa-wM>

2000 Nobel
Prize winner:

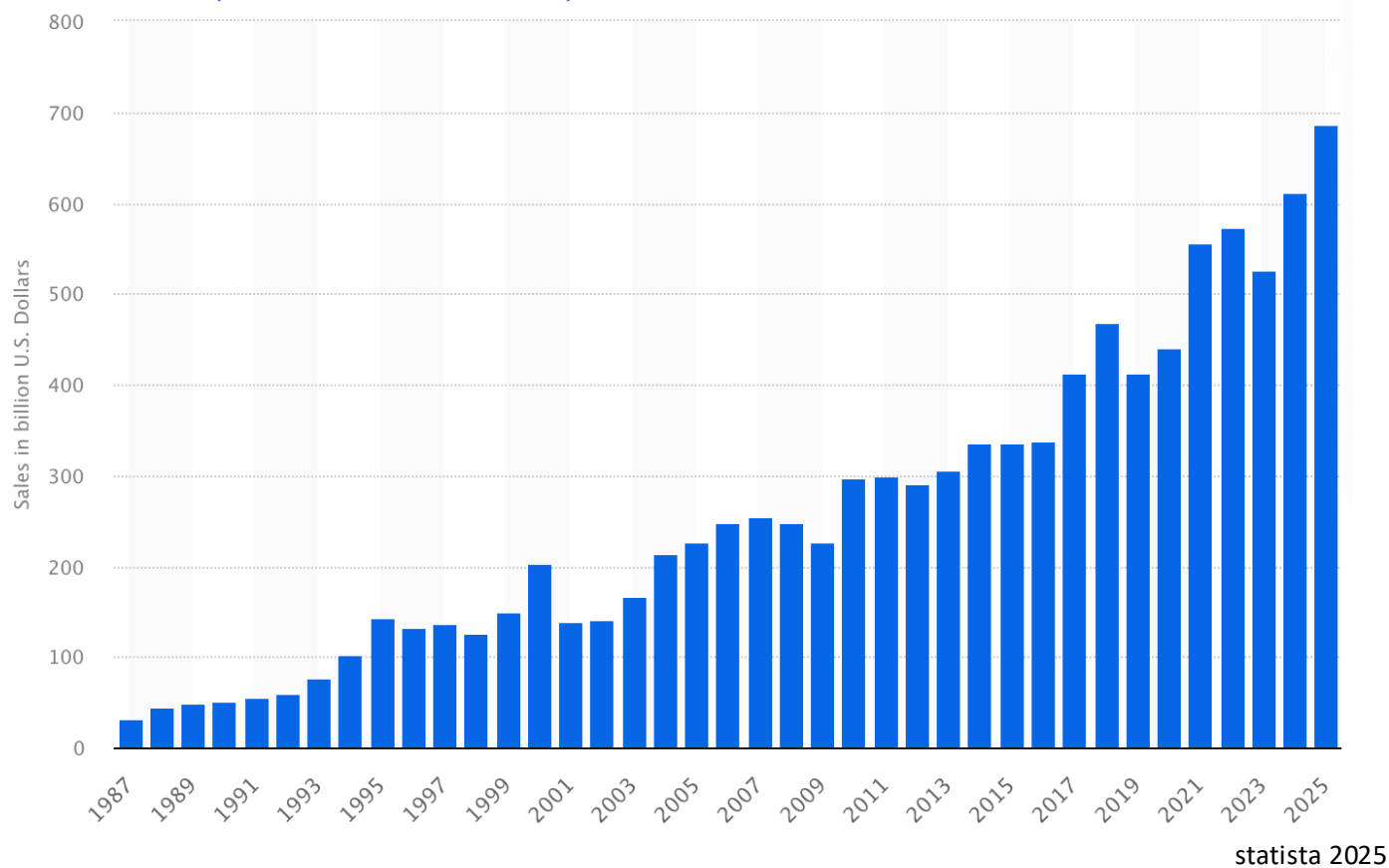


Later: Listen to Jack Kilby's Nobel lecture

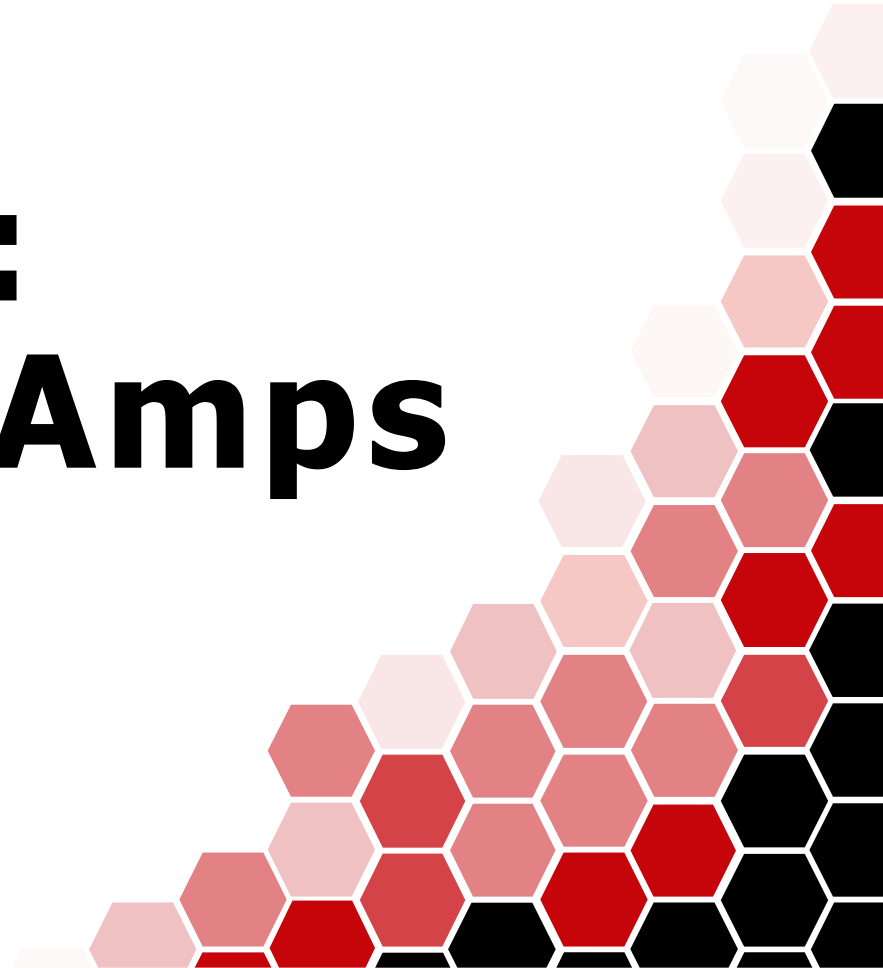
Why should I care?

And now it's a multi-trillion-dollar industry

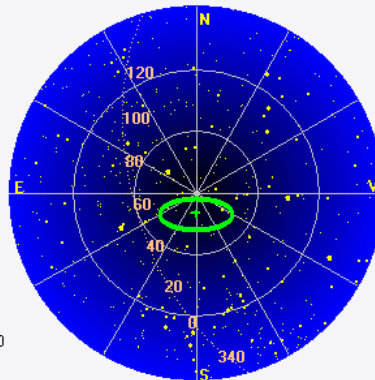
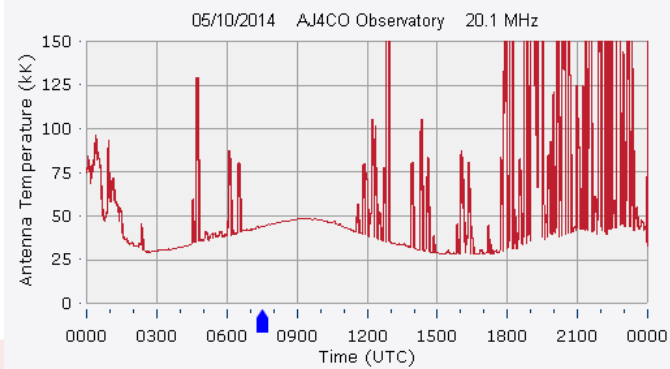
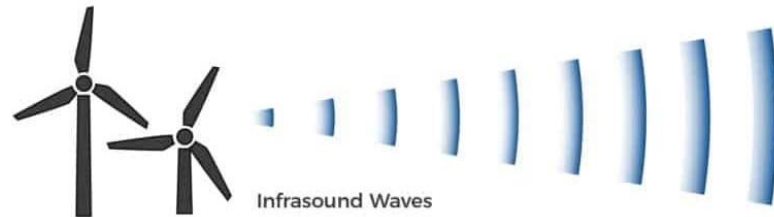
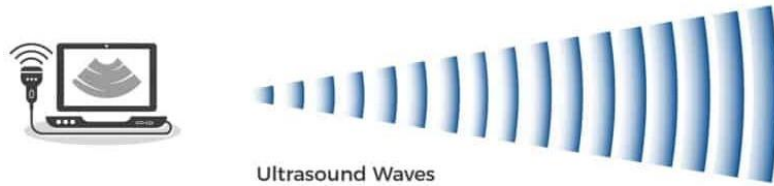
**Semiconductor market revenue worldwide from 1987
to 2025 (in billion U.S. dollars)**



Chapter 1: Signals & Amps



What is a signal?



NASA JOVE

Signals carry lots of information

- musical instruments carry audio info
- telescopes and cameras carry visual info
- the earth carries gravitational info
- air even carries thermal info!

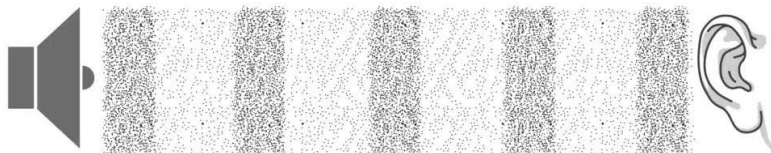
we want to process it all



How do we process signals?

Putting them in the electrical domain

With transducers! (transducer = “over” + “to lead”, or “to lead across”)



Our ears, eyes are transducers (e.g., eardrums)

What about with electronics?



The microphone is a transducer that transforms sound/pressure waves into an electrical signal

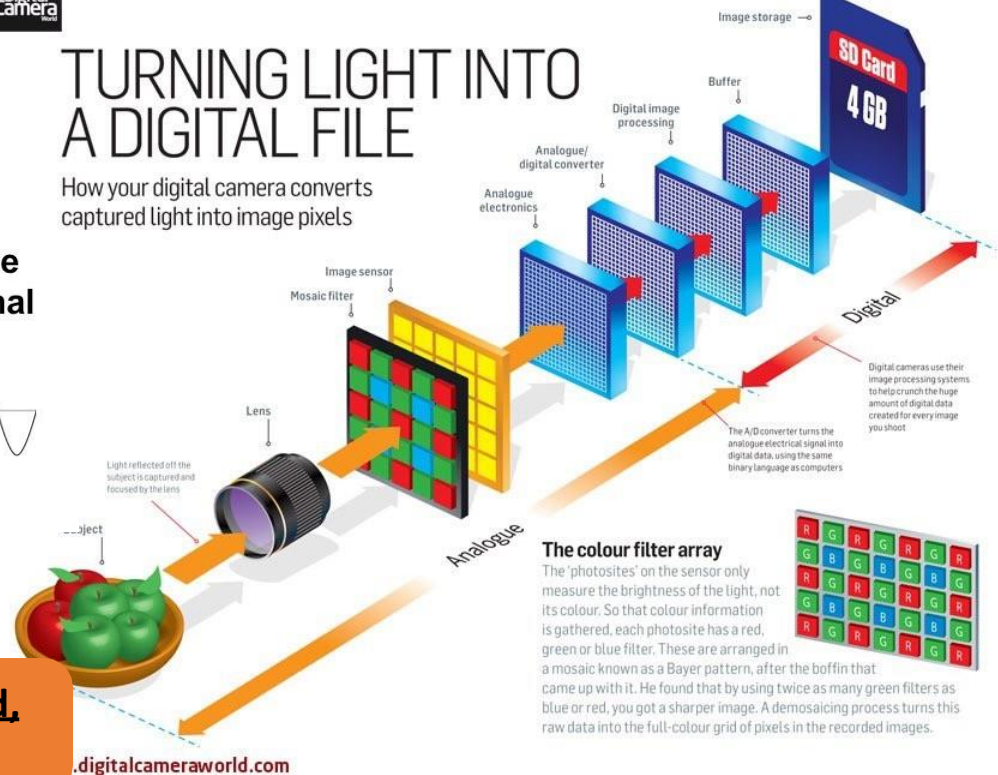
Once we have those signals converted, what do we do with them?

Your phone cameras contain an image sensor transducer that transforms light waves into electrical signals

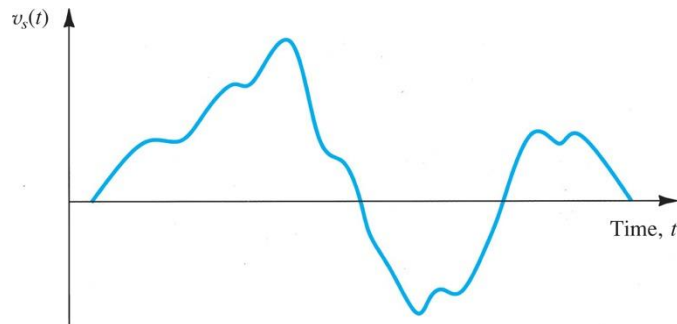
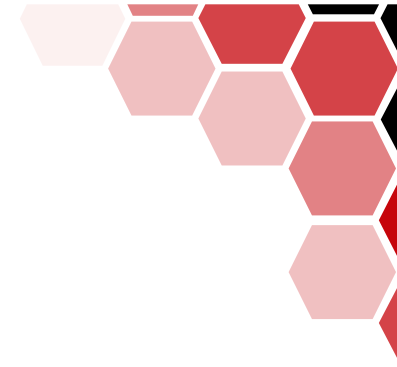


TURNING LIGHT INTO A DIGITAL FILE

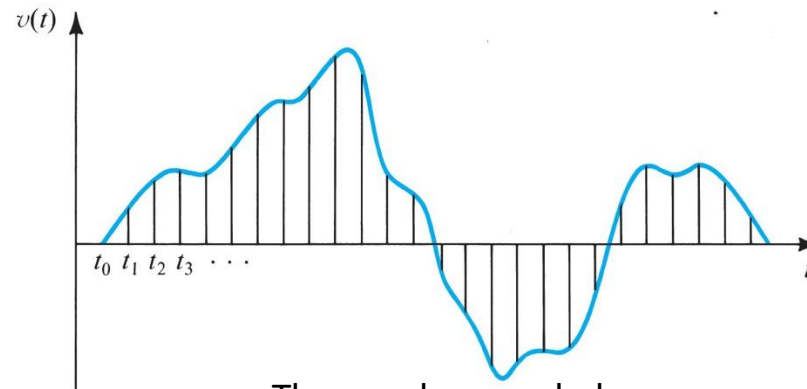
How your digital camera converts captured light into image pixels



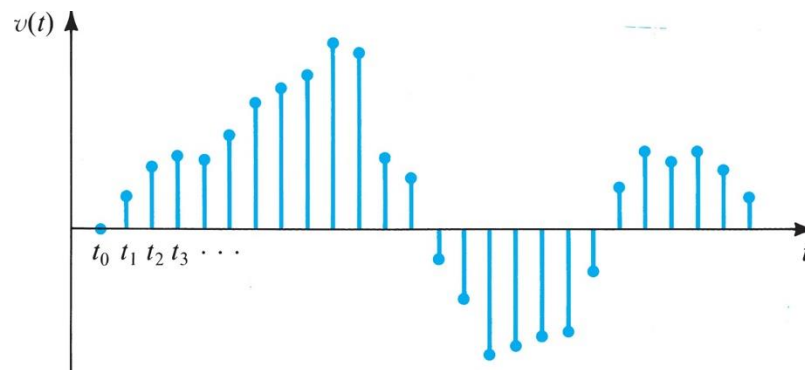
Analog, Digital, Periodic, oh my!



There are analog signals



They can be sampled



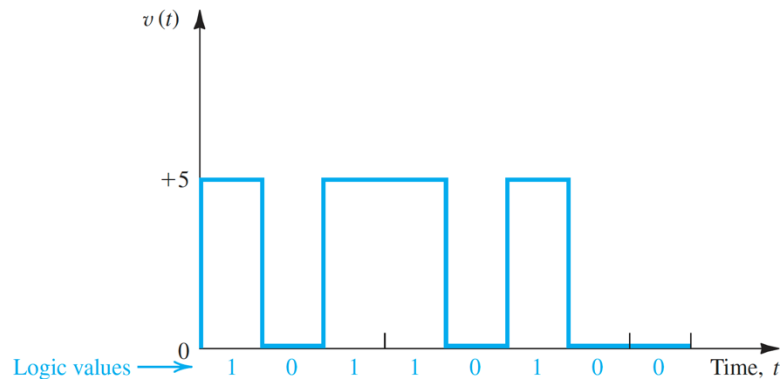
To give discrete-time signals

Signal information is compressed!

→ We lose information, but if there are enough samples we can reconstruct it accurately.



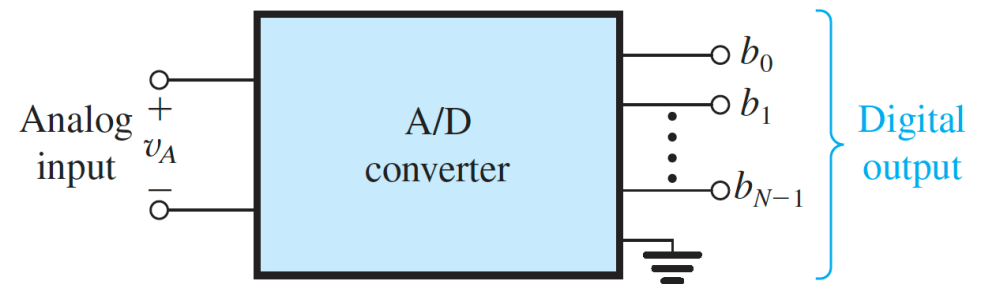
Analog, Digital, Periodic, oh my!



There are digital signals

They can be assigned logic values with total value of:

$$D = b_0 2^0 + b_1 2^1 + b_2 2^2 + \dots + b_{N-1} 2^{N-1}$$



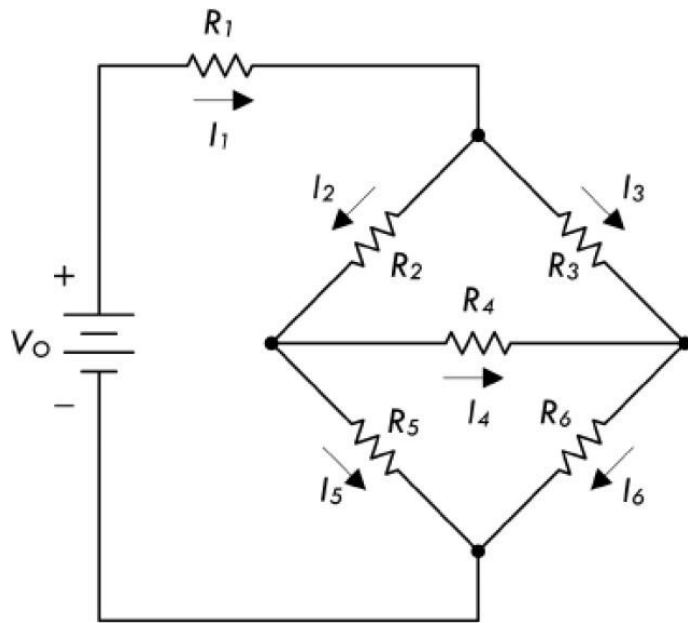
They can be converted from analog signals

8-, 10-, 12-, and 16-bit ADCs are common, but you can also have a 1-bit "ADC" (equivalent to a threshold function)

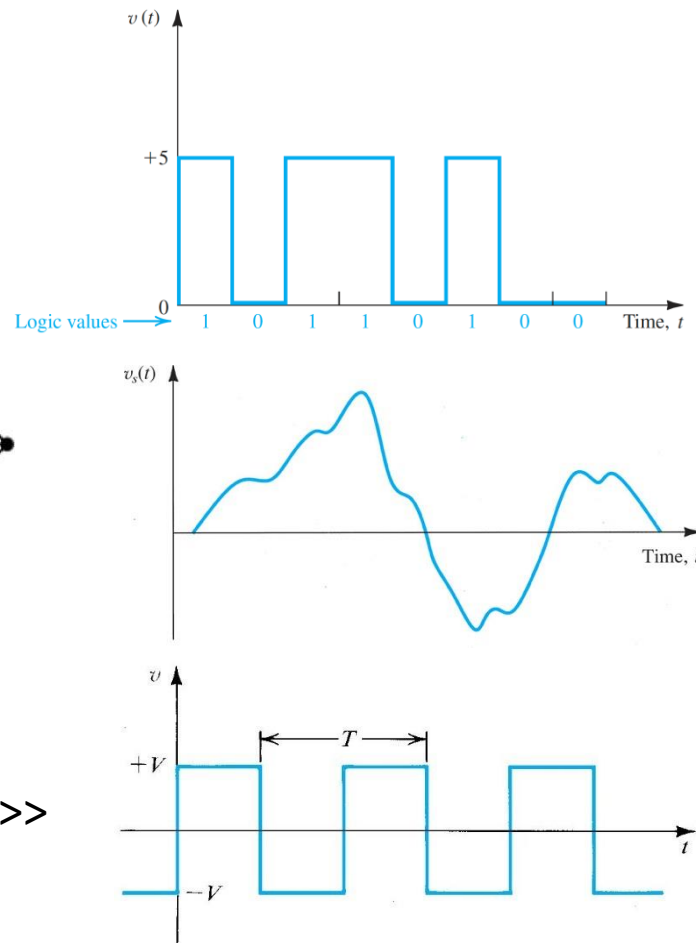
Many electronic systems contain both analog and digital components! They are called mixed-signal.

How do we process these signals?

With electronic circuits!



These are all time-varying >>
(transient analysis)



To build circuits, we need to understand how they behave at each point in time (DC analysis)

How do we process these signals?

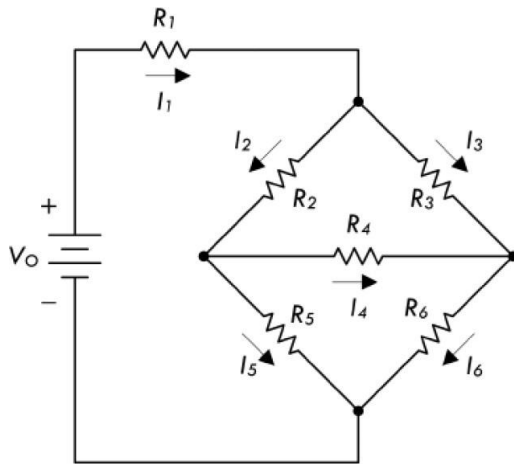
You may have left off with KCL and KVL in the last class. Here's a recap:

Kirchhoff's Current Law (or Junction Rule): The sum of the currents that enter a junction equals the sum of the currents that leave the junction:

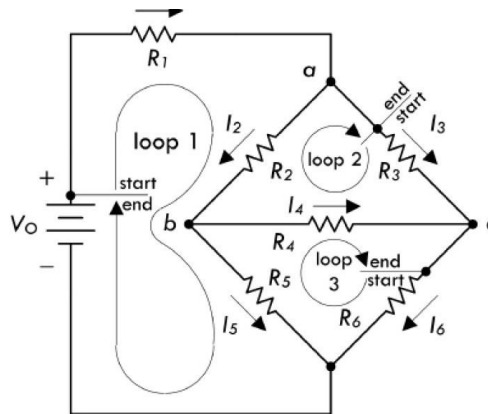
$$\sum I_{\text{in}} = \sum I_{\text{out}} \quad (2.21)$$

Kirchhoff's Voltage Law (or Loop Rule): The algebraic sum of the voltages around any loop of a circuit is zero:

$$\sum_{\text{closed path}} \Delta V = V_1 + V_2 + \dots + V_n = 0 \quad (2.20)$$



Example!



Equations resulting after applying Kirchhoff's current law:

$$I_1 = I_2 + I_3 \quad (\text{at junction a})$$

$$I_2 = I_5 + I_4 \quad (\text{at junction b})$$

$$I_6 = I_3 + I_4 \quad (\text{at junction c})$$

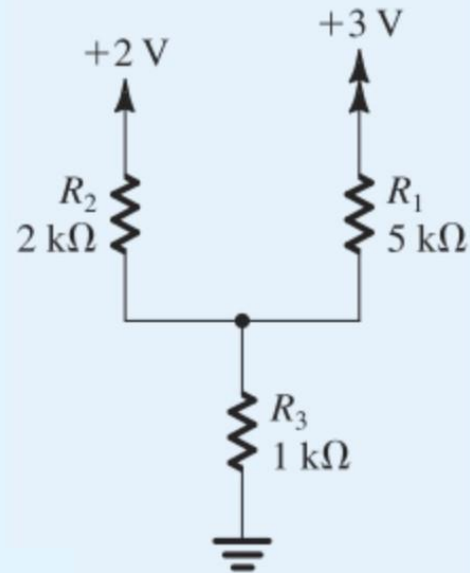
Equations resulting after applying Kirchhoff's voltage law:

$$V_0 - I_1 R_1 - I_2 R_2 - I_5 R_5 = 0 \quad (\text{around loop 1})$$

$$-I_3 R_3 + I_4 R_4 + I_2 R_2 = 0 \quad (\text{around loop 2})$$

$$-I_6 R_6 + I_5 R_5 - I_4 R_4 = 0 \quad (\text{around loop 3})$$

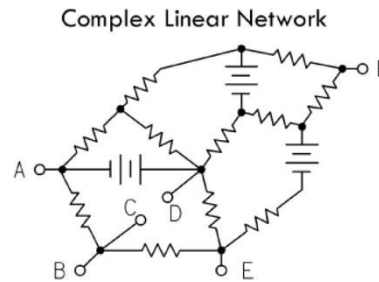
Example 1



- (a) Loop Equations: Define branch currents I_1 and I_2 in R_1 and R_2 , respectively; write two equations and solve them.
- (b) Node Equation: Define the node voltage V at the common node; write a single equation and solve it.

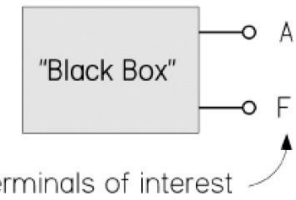
Two important theorems

Consider an example: when we connect our phones to a wall charger, it forms a complex network of components. We don't want to calculate how current flows!



We'd rather think of it as a lumped-together box, paying attention to input power + output power.

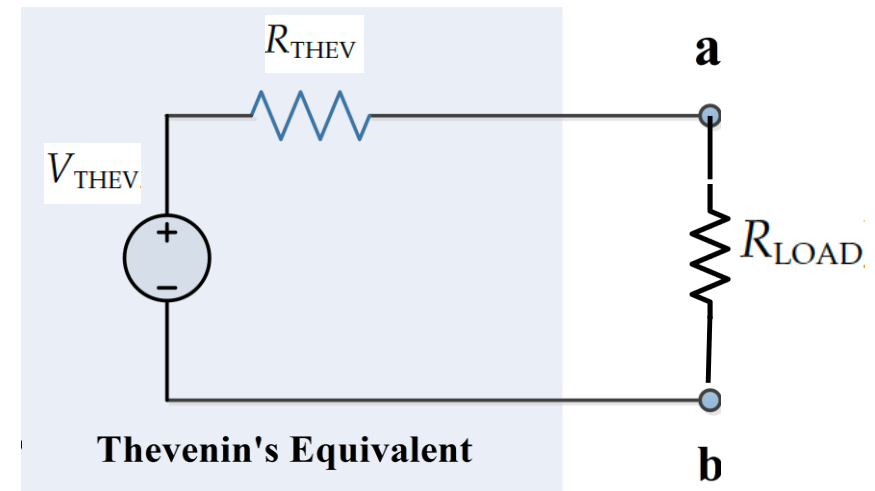
Black Box Analogy



We can simplify our sources using their equivalents:

Equivalent 1: Thevenin

Thevenin's theorem: Any linear two-terminal dc network can be represented by a voltage source in series with a resistor.



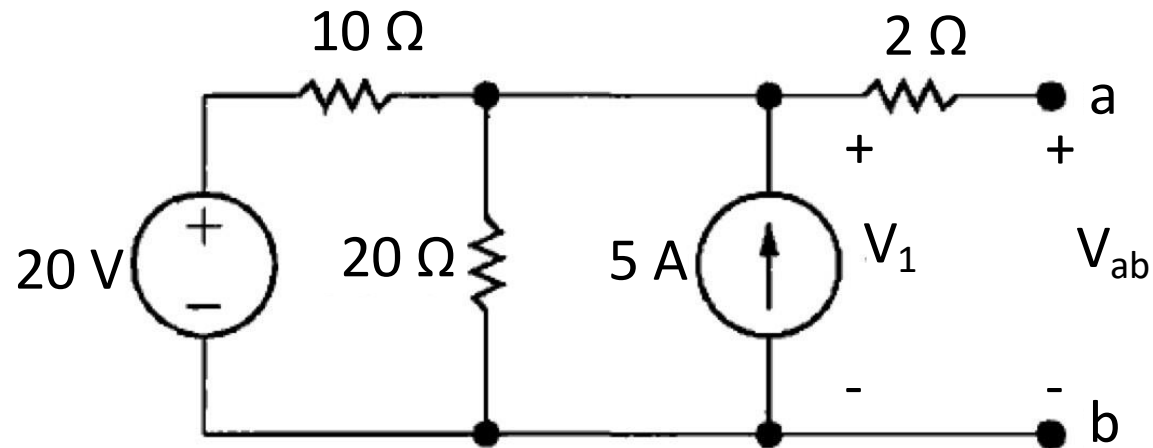
Output current: $I = V_{THEV} / (R_{THEV} + R_{LOAD})$

Example 2

Find the Thevenin equivalent

Step 1: Find V_{THEV} by removing load, finding open-circuit voltage.

Step 2: Zero sources out and calculate R_{THEV} .
Equivalent resistance is across a and b.



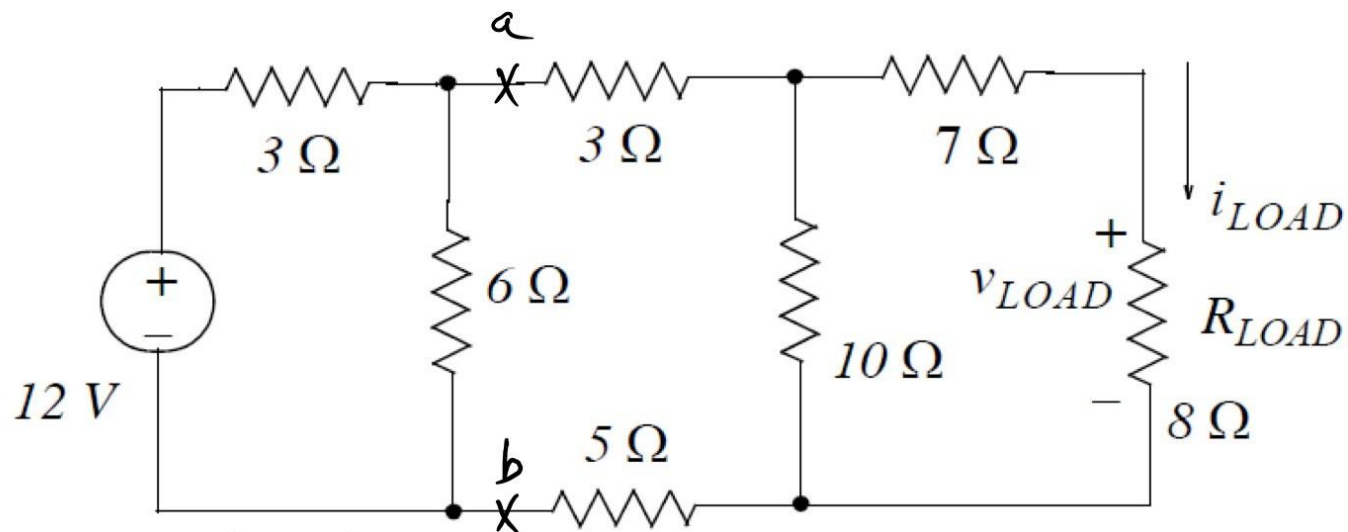
Note: $I = V_{\text{THEV}} / (R_{\text{THEV}} + R_{\text{LOAD}})$

Example 3

Step 1: Find V_{THEV} by removing load, finding open-circuit voltage.

Step 2: Zero sources and short the load to calculate R_{THEV} . Equivalent resistance is across nodes of choice.

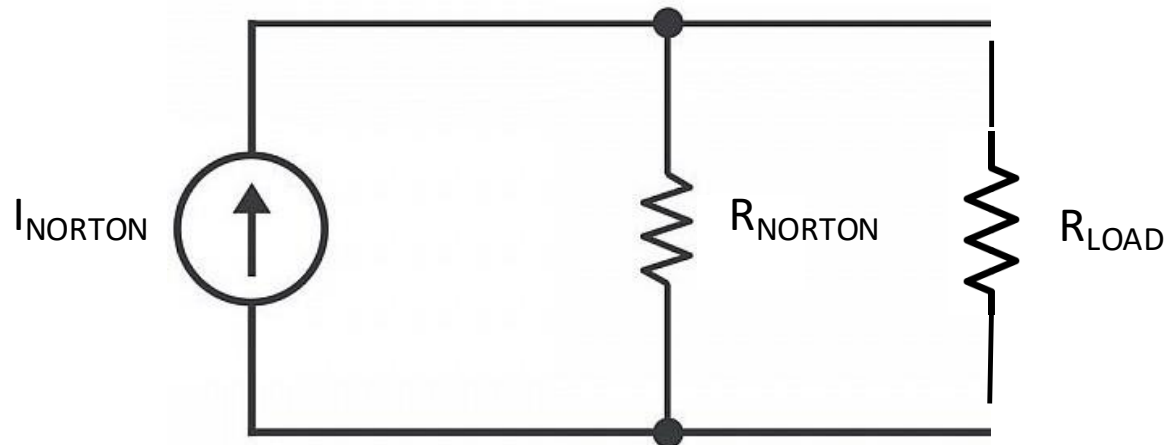
Find i_{LOAD} through R_{LOAD} using Thevenin's theorem



Two important theorems

Equivalent 2: Norton

Norton's theorem: Any linear two-terminal dc network can be represented by a **current source** in parallel with a resistor.



To convert from Thevenin equivalent:

Step 1: Find I_{NORTON} by replacing load with a short.

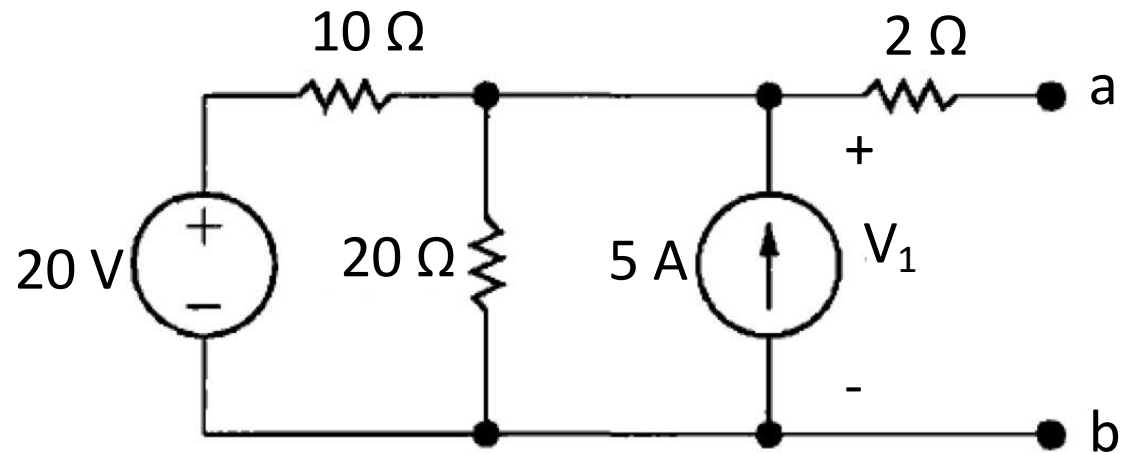
Step 2: Zero the sources and open R_{LOAD} to get R_{NORTON} .

Example 4

Find Norton equivalent

Step 1: Find I_{NORTON} by replacing load with a short.

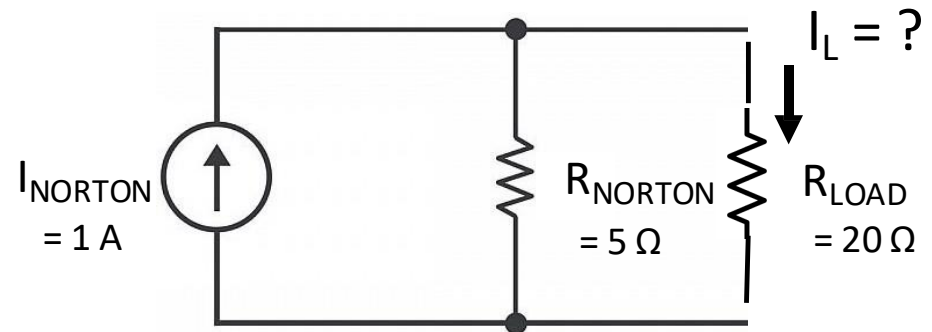
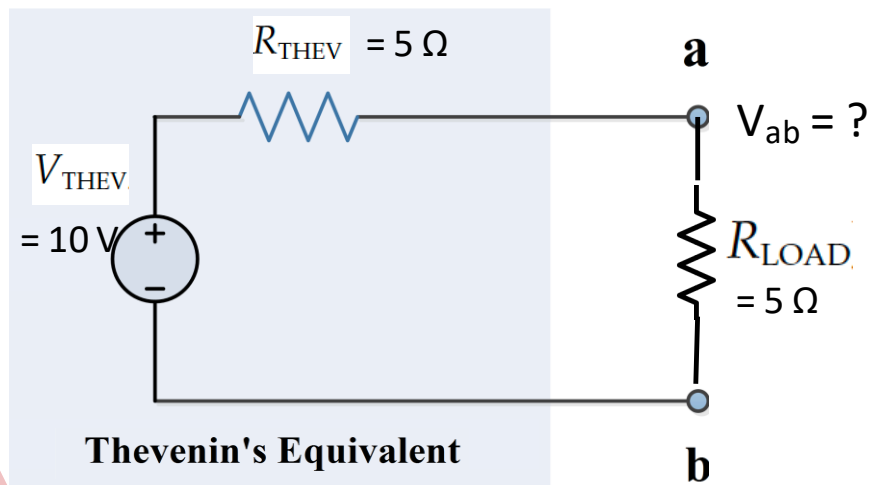
Step 2: Zero the sources and open R_{LOAD} to get R_{NORTON} .



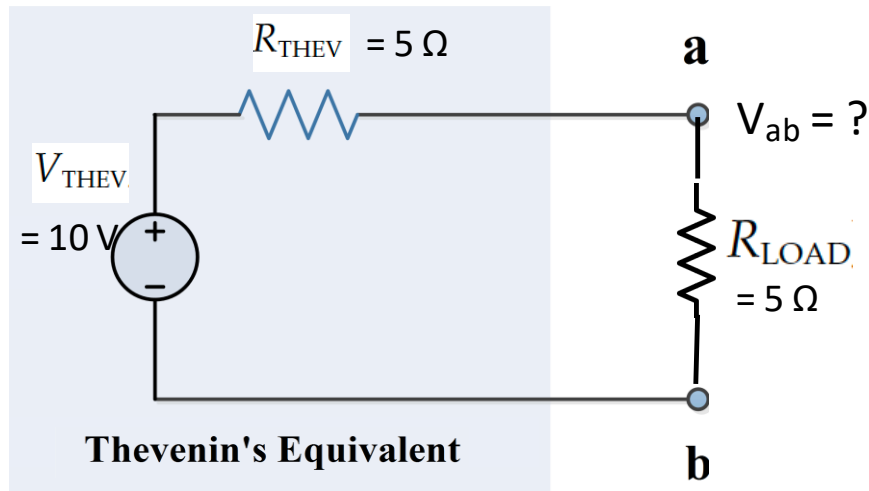
An important note

Although you know what your V_{THEV} and I_{NORTON} values are, the **DELIVERED Voltage/Current to R_{LOAD} can be DIFFERENT** depending on the ratio between $R_{\text{THEV}}/R_{\text{NORTON}}$ and R_{LOAD} .

Find V_{ab} and I_L in the following examples to see this in practice!



Output voltage and current differs

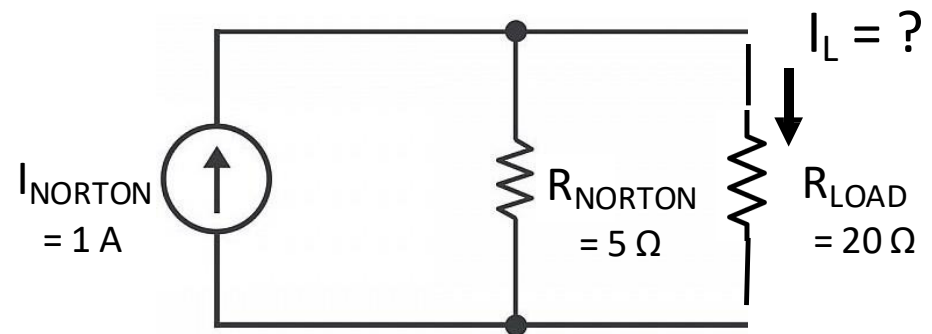


Method 1:

$$I = 10 \text{ V} / (5 \Omega + 5 \Omega) = 1 \text{ A}$$

$$V_{ab} = 5 \Omega * 1 \text{ A} = 5 \text{ V}$$

2 steps!!



Method 1:

$$V = I * (5 \Omega || 20 \Omega) = 1 \text{ A} * (5 * 20 / (5 + 20)) = 4 \text{ V}$$

$$I_L = 4 \text{ V} / 20 \Omega = 1 / 5 \text{ A} = 0.2 \text{ A}$$

Method 2 - dividers

A very common circuit element you will see time and time again >>>

Because the current is

$$I = \frac{V_{in}}{R_1 + R_2}$$

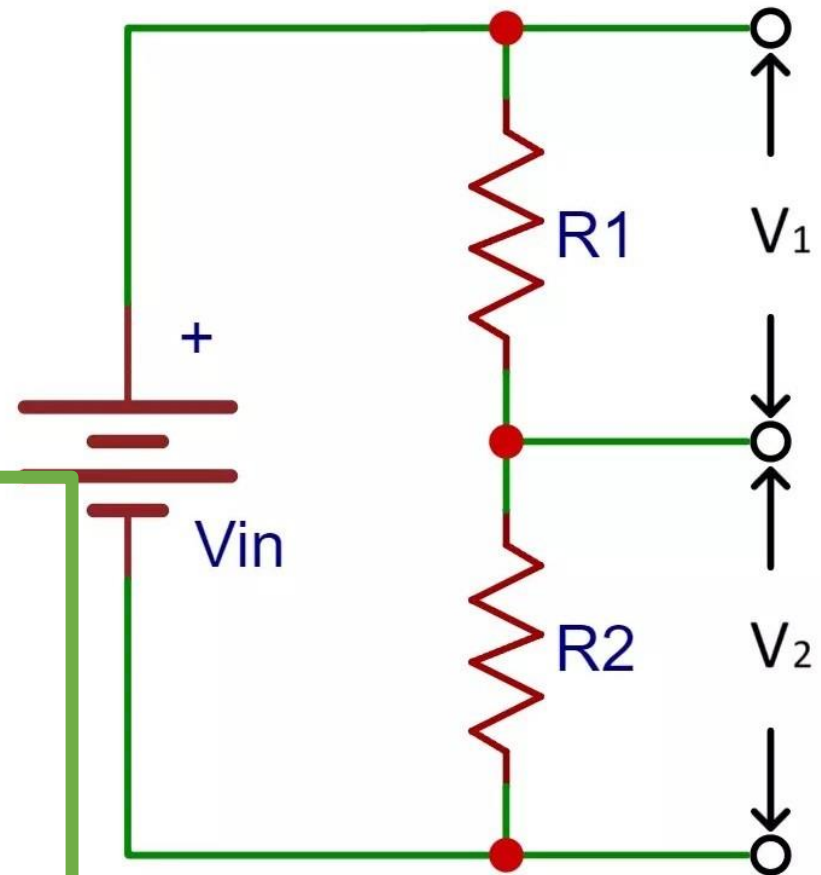
Then

$$V_1 = I * R_1 = V_{in} * \frac{R_1}{R_1 + R_2}$$

And

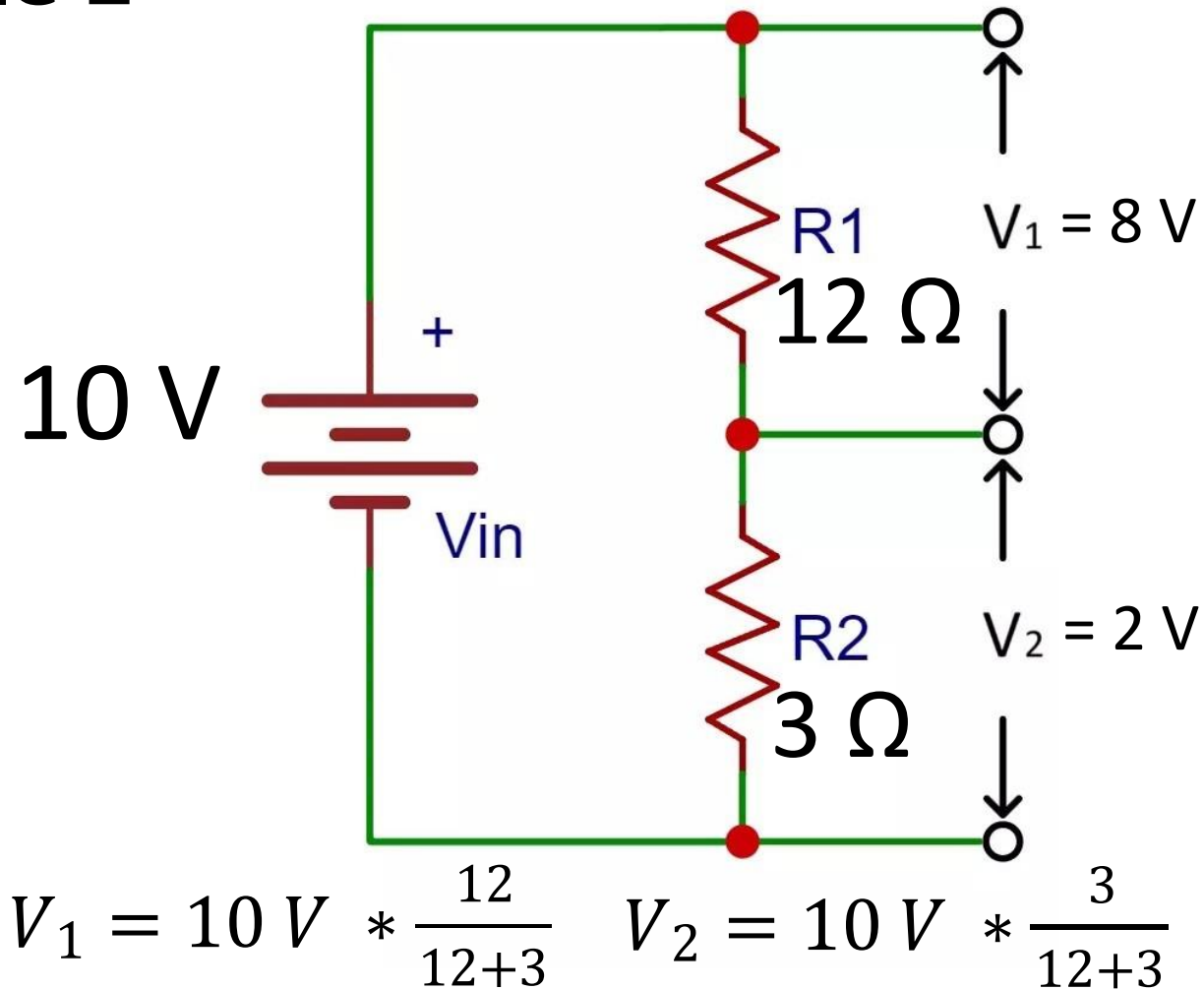
$$V_2 = I * R_2 = V_{in} * \frac{R_2}{R_1 + R_2}$$

Voltage divider



<<Know this by heart!

Example 2



One step!!

Method 2 - dividers

Another very common circuit element
you will see time and time again >>>

Because the voltage is

$$V_{in} = I_{in} \frac{R_1 R_2}{R_1 + R_2}$$

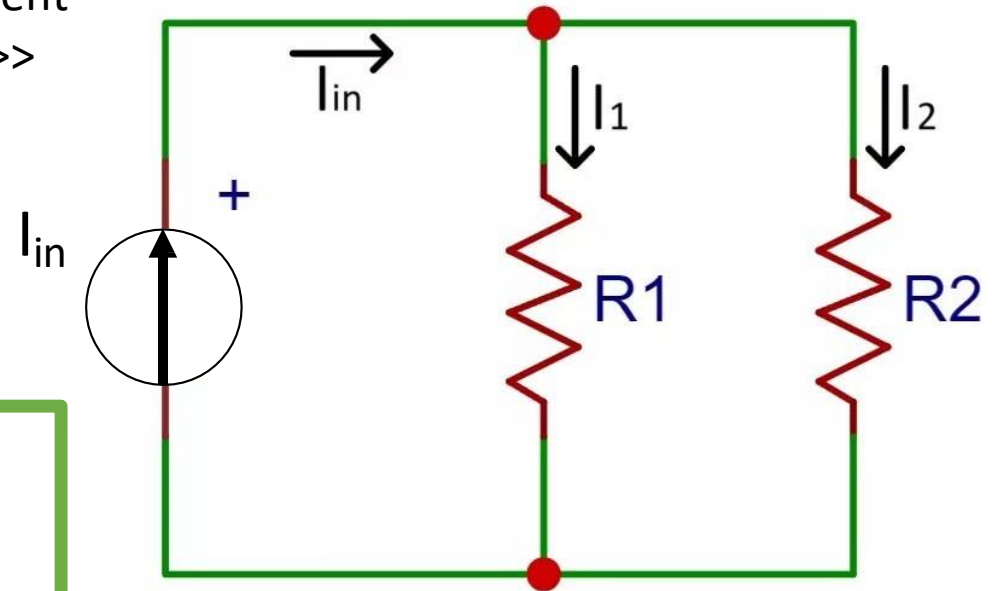
Then

$$I_1 = \frac{V_{in}}{R_1} = I_{in} * \frac{R_2}{R_1 + R_2}$$

And

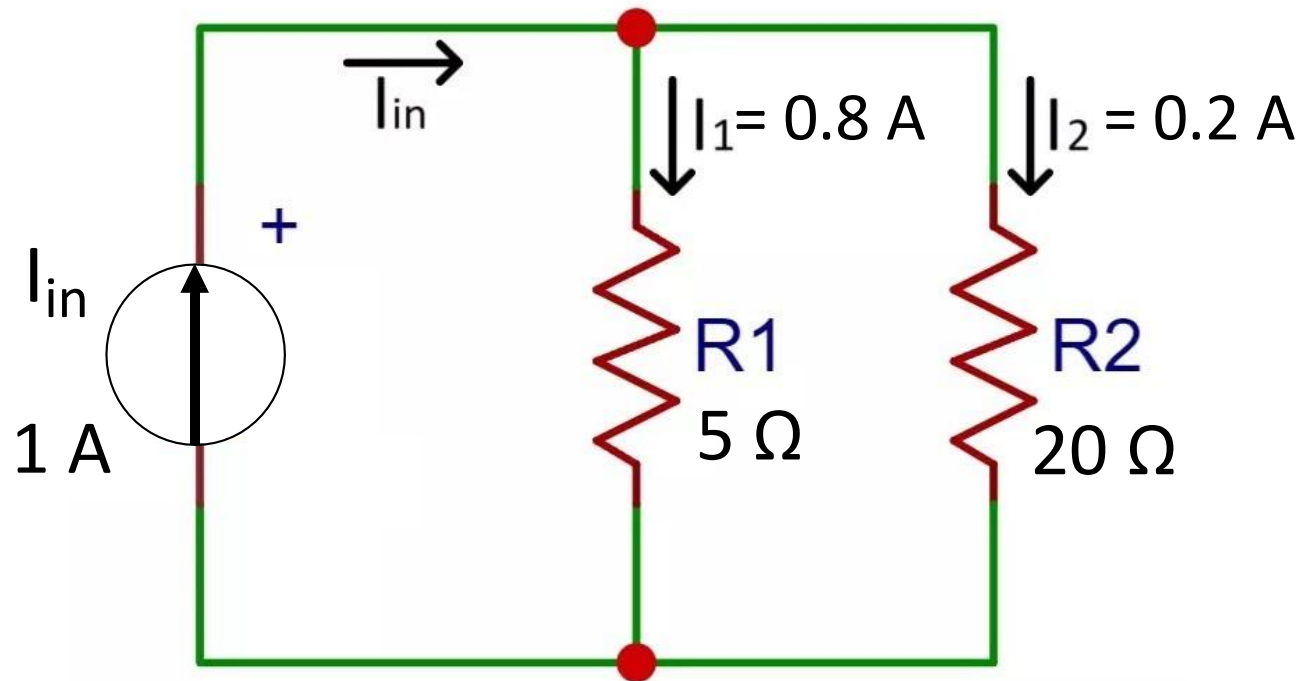
$$I_2 = \frac{V_{in}}{R_2} = I_{in} * \frac{R_1}{R_1 + R_2}$$

Current divider



<<Know this by heart!

Example 3



$$I_1 = 1\text{ A} * \frac{20}{5+20} \quad I_2 = 1\text{ A} * \frac{5}{5+20}$$

1 step!!

Questions?

**See you next
time!**

